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Wireless Security Report - GNSS

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1 INTRODUCTION

Global Navigation Satellite Systems *(GNSS)* are fundamental to modern location services and are heavily relied upon by smartphones to calculate positioning. However, despite their widespread use, GNSS signals share a critical vulnerability: **spoofing**. This is an attack where counterfeit signals deceive the receiver into generating false location or timing data.

This study investigates the GNSS measurement patterns of a consumer-grade Android smartphone. To establish a baseline of performance, raw GNSS data was collected and processed in a real-world urban environment near the *Politecnico di Torino*. The data collection involved two distinct scenarios:

- A normal session, recorded under standard operating conditions.
- An interfered session, recorded with an active phone call taking place nearby.

Following this initial baseline analysis,a spoofing attack was simulated by artificially injecting fake coordinates and time delays into the system to observe the device's response.

The remainder of the report is structured as follows:

- **Section 2:** Details the experimental setup, including the smartphone configuration, the MATLAB environment, and the data processing pipeline.
- **Section 3:** Analyzes the results, comparing the different recording scenarios and evaluating the overall impact of the spoofing simulation.
- **Section 4:** Summarizes the key findings and provides concluding remarks on the experiment.

2 METHODS

2.1 Devices and Softwares

To conduct the experiment a **Samsung Galaxy S25** running *Android 16* was used. Data logging was performed using *GNSS Logger v2.0.0.2*.

For data analysis, *MATLAB R2025b* ([5]) was chosen, as it provides the capability to process raw GNSS measurements using the provided script to accurately compute the Position, Velocity, and Time *(PVT)* solution.

2.2 Data Collection Procedure

Two distinct 5-minute GNSS data logging sessions were conducted at different times but in the exact same location (specifically, the atrium area immediately outside the classroom, next to the R3 building (Fig. 1)) to evaluate the impact of local RF interference on receiver performance.

The **first session** took place outdoors on March 18, 2026, at approximately 17:10, representing a baseline scenario without interference.

The **second session** was conducted on March 25, 2026, at approximately 16:20, in the same outdoor location, introducing an active phone call as a source of interference.

Both sessions utilized the **GNSS Logger app** configured with the following technical settings:

- *GNSS Location*: to capture location data.
- *GNSS Measurements*: to log raw GNSS measurements.
- *Navigation Messages*: to capture navigation data.
- *GnssStatus*: to log GNSS status information.
- *Sensors*: to capture sensor data.

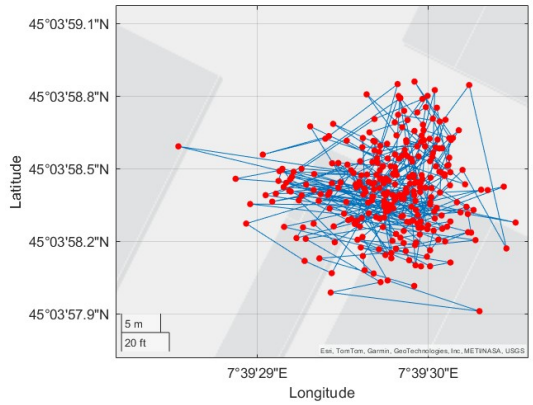

Figure 1: Data Collection Map

To verify the consistency of the satellite geometry between the two acquisitions, the corresponding skyplots and satellite distributions are provided in Appendix A (Fig. 10).

2.3 Baseline Scenario

During the baseline experiment, the smartphone remained stationary. The measurement setup featured a single building in front, with a free sky view behind and on the sides, representing nominal outdoor conditions with minimal signal obstruction.

2.4 Interference Scenario

For the interference scenario, the measurement session utilized the exact same physical setup and configuration as the baseline experiment. Throughout the data logging process, the target smartphone

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was engaged in an active voice call with another nearby device to introduce the required local interference.

2.5 Spoofing Setup

Spoofing scenarios were **mathematically emulated** during post-processing using the *Google's GPS measurement tool* in MATLAB, injecting artificial variations directly into the raw baseline measurements. The following script parameters were manipulated to simulate a spoofing attack:

- `spoof.active`: Set to 1 to enable the attack.
- `spoof.position`: Assigned modified coordinates to emulate a counterfeit location (`45.076795, 7.683674`).
- `spoof.delay`: Adjusted in milliseconds to mimic spoofer-to-victim propagation delays.

3 RESULTS AND DISCUSSION

3.1 Baseline scenario

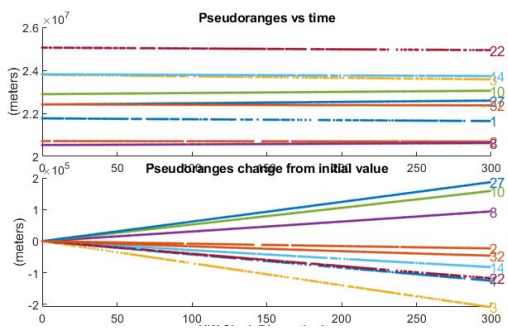


Figure 2: Pseudorange Evolution

3.1.1 Pseudorange Evolution. Figure 2 displays code **pseudorange measurements** for tracked satellites during the baseline session, with absolute values $2.30 \cdot 10^7$ m (top subplot) consistent with *GPS L1* geometry from 20.180 km orbital radius. This suggests **stable signal** acquisition under favorable conditions.
The middle subplot shows pseudorange variations over 300 seconds, with absolute changes reaching up to $\pm 2.0 \cdot 10^5$ m. These large linear shifts are expected and reflect **normal high-speed** orbital motion, clearly distinguishing satellites moving away (positive slopes, e.g., *PRN 27*) from those approaching the receiver (negative slopes, e.g., *PRN 3*).

3.1.2 Pseudorange Rate. Validating GNSS data quality involves verifying the internal consistency of the receiver's measurements. The recorded radial velocities reflect expected orbital dynamics, ranging from approximately -800 to $+600$ m/s. To validate these values, the empirical time **derivative of the raw pseudoranges** (the physical change in measured distance over time) is compared against the **instantaneous pseudorange rate** directly reported by the smartphone. The analysis confirms a **good overall tracking stability**: the mean differences between these two independently computed metrics are really low, falling between -0.90 and 0.87 m/s. This sub-meter per second alignment demonstrates highly reliable signal tracking and robust synchronization within the receiver's internal processing loops during the static baseline scenario.

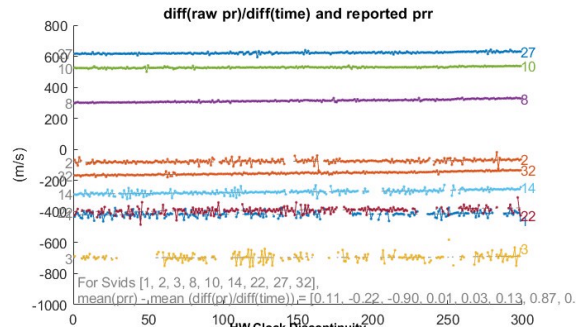


Figure 3: Pseudorange Rate

3.2 Spoofed version of the baseline

For the spoofed test, the coordinates of **Piazza Castello** were manually injected (`45.076795, 7.683674`) directly into the processing script at time $t = 50s$ of the simulation. Comparing the result with the baseline, it can be observed how the position undergoes an instantaneous deviation, stabilizing on the spoofed target as shown in the Geoplot. Since the mathematical algorithm estimates the

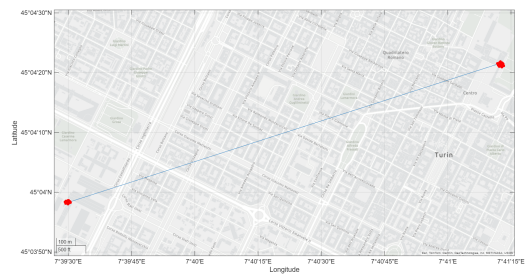


Figure 4: Map of the spoofed scenario

PVT solution without inertial filtering, an effective detection strategy involves monitoring for kinematic anomalies in the derived observables. Analysis of the WLS states (Figure 6) confirms the instantaneous nature of the spoofing attack, evidenced by a sharp step-function in the position offsets at $t = 50$ s. Although this positional jump induces an initial velocity disturbance, the maximum kinematic violation occurs later, at $t = 72$ s. At this epoch, the horizontal velocity module peaks at 2.36 m/s, accompanied by a severe anomaly in the vertical velocity component (Down state), which drops to nearly -2 m/s. These spikes demonstrate that the instantaneous coordinate shift creates a profound geometric instability within the estimator. Combining these observations with a standard deviation $\sigma_{spoofed} = 0.226$ m/s (which is 38% greater than that of the baseline $\sigma_{baseline} = 0.163$ m/s) provides a highly robust signature for detecting spoofing attacks.

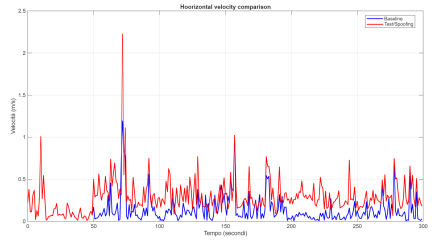


Figure 5: Horizontal velocity comparison

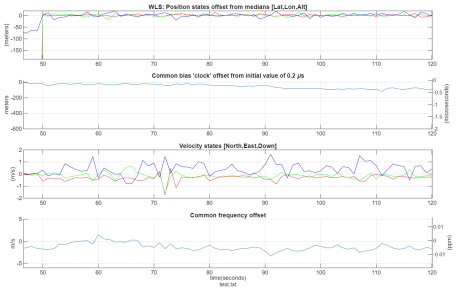


Figure 6: WLS and velocity anomalies

Subsequently, the cyberattack was repeated by introducing an artificial time delay of 0.001 seconds (spoof.delay) into the spoofer without altering the target coordinates. In this scenario, the estimated 3D position of the receiver does not change significantly. However, significant deviations are noticeable not only in the raw pseudorange plots but, more importantly, in their first derivative, diff(raw)/diff(time). The sudden injection of the time delay forces the algorithm to process an instantaneous jump in the pseudoranges, creating a massive spike in the derivative plot. The reason the spatial coordinates (x, y, z) remain unaffected lies in the mathematical formulation of the navigation equations. Because the artificial delay is applied uniformly across all tracked satellite channels, the system interprets it as a common expansion of all distances. Consequently, the Weighted Least Squares (WLS) estimator absorbs this uniform error entirely into the Time state, which is clearly evidenced by the sudden offset in the receiver's internal clock bias. The consequences of this specific attack on a GNSS receiver's performance depend heavily on its application. From a pure navigation perspective, an ordinary smartphone user would not notice any anomaly, as the physical location on a map would appear uncompromised. However, at the hardware level, the receiver's performance is severely degraded in the Time domain of the PVT solution.

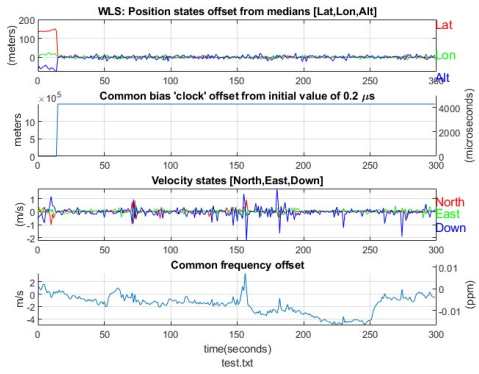
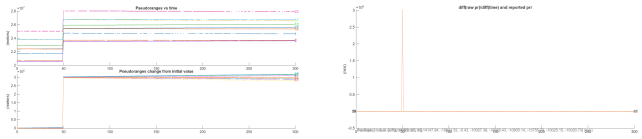


Figure 7: Clock bias anomaly

Inside the smartphone, introducing a sudden millisecond error in the GNSS time can disrupt the synchronization between the GNSS chip and other internal sensors, potentially causing internal failures in specific applications. On a broader scale, if this same timing attack were directed at receivers used in critical infrastructures (such as 5G telecommunication networks or others which rely on precise GNSS time synchronization rather than positioning) this invisible desynchronization would cause catastrophic system failures. Therefore, monitoring the derivative spike and the consequent sudden offset in the clock bias serves as a critical defense mechanism against timing attacks.



(a) Pseudoranges vs Time **(b)** diff(raw pr)/diff(time)

3.3 Voice call interference scenario

In the voice call scenario, most of the analyzed parameters show results congruent with the baseline experiment. However, there is a significant deviation in the altitude component of the Position State. Unlike latitude and longitude, which behave as expected, the altitude graph shows severe instability, characterized by significant spikes dropping up to minus 60 meters. This is easily noticeable in the latter part of the data collection. Other metrics behave normally: velocity and common frequency offset stabilize near zero, consistent with a static receiver. The clock bias also shows results close to the baseline experiment. Overall, the call interference doesn't seem to create visible effects other than this severe degradation in altitude.

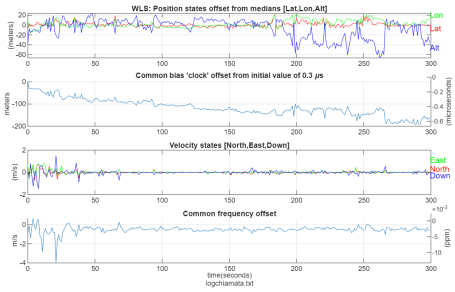


Figure 9: WLS anomalies: voice call scenario

4 CONCLUSION

This study evaluated **smartphone GNSS performance** under nominal, interfered, and spoofed conditions. While the **baseline** demonstrated stable tracking, **local RF interferences** significantly degraded positional accuracy, highlighting consumer hardware sensitivity. Crucially, software-injected spoofing successfully manipulated the position solution without altering standard metrics like $C/N0$ or $HDOP$. However, the **unnatural absence of thermal noise** in the spoofed pseudoranges provides a clear mathematical **anomaly** for detection.

Future work should explore physical spoofing via *Software-Defined Radios (SDR)* and develop algorithms that exploit these signal variance anomalies to identify deceptive transmissions.

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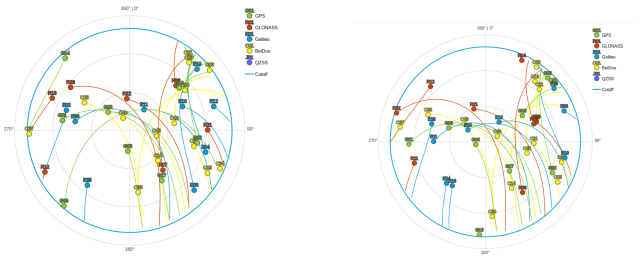
A APPENDIX

A.1 Satellite Geometry and Sidereal Day Shift



(a) Baseline scenario. **(b)** Interference scenario.

Figure 10: Satellite geometry and expected trajectories generated via GNSS Planning [1].

Fig. 10b illustrates the **satellite constellations** during the baseline (a) and interference (b) scenarios, recorded 7 days apart. While the orbital ground tracks (solid lines) perfectly overlap, a slight displacement in the instantaneous satellite positions (dots) is observable. This occurs because the GNSS satellite geometry repeats every sidereal day (approximately 23 hours and 56 minutes), advancing by about 4 minutes each solar day. After exactly 7 days, the identical constellation appears 28 minutes earlier. Since the baseline was recorded at 17:10 and the second session at 16:20, there is a 12-minute time difference relative to the exact sidereal alignment, which physically accounts for the minor shifts along the visible trajectories.

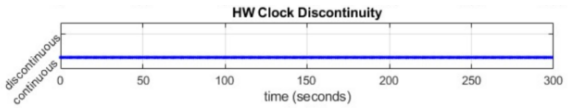


Figure 11: Hardware Clock Discontinuity state over the 300-second acquisition period.

Fig. 11 illustrates the state of the receiver's hardware clock throughout the 300-second data logging session. The plot displays a steady, continuous line, indicating that the smartphone's GNSS chip did not experience any hardware clock discontinuities. This behavior confirms that the Android operating system **did not trigger battery management** mechanisms that would otherwise turn off the user clock to save power.

As a result, the absence of these hardware interruptions prevents a growing user clock bias with respect to the GNSS time , thereby avoiding sudden jumps in the raw pseudorange measurements and maintaining the overall stability of the calculated Position, Velocity, and Time (PVT) solution.